Aiden Fox

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Education

University of Southern California (USC) — Los Angeles, CA Aug 2024 – Dec 2027

BS Computer Engineering & Computer Science, GPA 3.61

Leadership: E-Board Theta Tau, E-Board Beta Theta Pi (Rush Chair), Treasurer of Quantum Engineering and Ethics.

Experience

ML / Fullstack Software Engineer Intern, Motif Inc. — Remote May 2026 – Present

- Architected a deep-research platform feature from 0 to 1, taking it from a bare idea into a model-agnostic, reusable framework that downstream product features compose on to structure their outputs: one tool interface over two interchangeable provider backends, a thin base skill plus domain skills, and 24 mocked per-provider tests.
- Authored five reusable agent skills on that framework — product discovery, vendor sourcing, building-code compliance research, model summarization, and floor-plan generation — including a code-research skill spanning eight regulatory domains and six international code families under a stricter citation contract than the tools it composes.
- Built product-card generation so deep research returns validated structured products instead of prose: every link is checked before display, a dead vendor page nulls that field rather than shipping a broken card, and cards persist across a refresh; the persistence layer's concurrency retry dropped 3 of 8 concurrent writes until an advisory lock took it to 0 dropped at 32 writers.
- Architected an LLM deep-research platform feature from 0 to 1 into a model-agnostic, reusable framework that downstream product features compose on: one tool interface over two interchangeable provider backends, a thin base skill plus domain skills, and 24 mocked per-provider tests.
- Root-caused a Kubernetes/Helm deployment gap that had silently broken every LLM-backed workflow node in one environment: proved from the live cluster that the async worker had no LLM API credentials and a memory limit 13x too small, shipped the fix, and added a config-preflight test so the class of gap fails loudly in CI.
- Designed a typed tool-provider registry so each workflow run receives only the tools its own spec declares and name collisions fail loudly; 30 saved workflows by 11 authors run through that path.

Software Engineer, Contract, Energy Connect — Remote May 2026 – Present

- Shipped ATOM, an in-app AI assistant whose capability map is generated from the live API surface and guarded by a CI drift check, so the agent's knowledge of available actions cannot silently fall out of sync with the application.
- Closed 842 tracked issues across four customer portals of a multi-tenant logistics platform.

Projects

GitSpider, CI Monitoring SaaS Bun, Hono, Postgres, Stripe, Fly.io

- Built and launched a solo production SaaS monitoring GitHub Actions CI, detecting p95 runtime regressions, flaky tests, and wasted runner minutes, then commenting the regression and blame commit on the pull request.

Technical Skills

Languages: C, C++, Python, Java, JavaScript, TypeScript, Verilog

AI Agents & LLM Systems: LLM tool-calling and agent runtimes, OpenAI + Anthropic APIs, prompt engineering and evaluation, A/B prompt experiments

ML & AI: PyTorch, HuggingFace Transformers, vision-language models (Qwen2.5-VL, SmolVLM), CLIP embeddings

Embedded & Hardware: Cortex-M7 firmware, FPGA (Xilinx, Vivado), RTOS, KiCad, PCB bring-up, I2C/SPI/UART, logic analyzers, oscilloscopes

Robotics & Real-Time: ROS2, micro-ROS, LIDAR-Inertial SLAM, Jetson, ISR control loops, watchdogs and safety FSMs

Cloud & DevOps: AWS (Lambda, SQS, SageMaker, EC2, S3), Docker, Kubernetes, Helm, GitHub Actions, GitLab CI/CD